

Chapter 132

HDR Palette

This chapter focuses on the global and zone-based adjustments found in the HDR palette, which is DaVinci's newest primary grading tool. Despite its name, this palette can be used, as you might the Lift/Gamma/Gain mode of the Primaries palette, to create the foundational adjustment for any SDR or HDR grade.

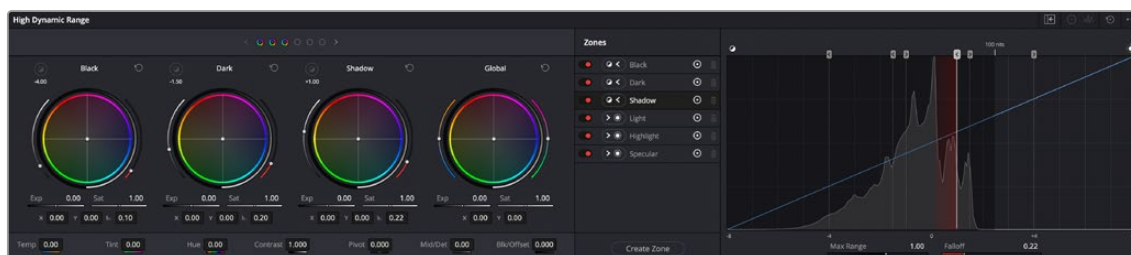
However, its zone-based nature gives it the power and specificity of curve-based adjustments, making the HDR palette suitable for a multitude of creative and corrective tasks. Furthermore, this palette's color space awareness, as well as the perceptually uniform color space in which its adjustments are made, make this palette uniquely suited for controlling the saturation and highlights of HDR mastered images.

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Introduction to the HDR Palette

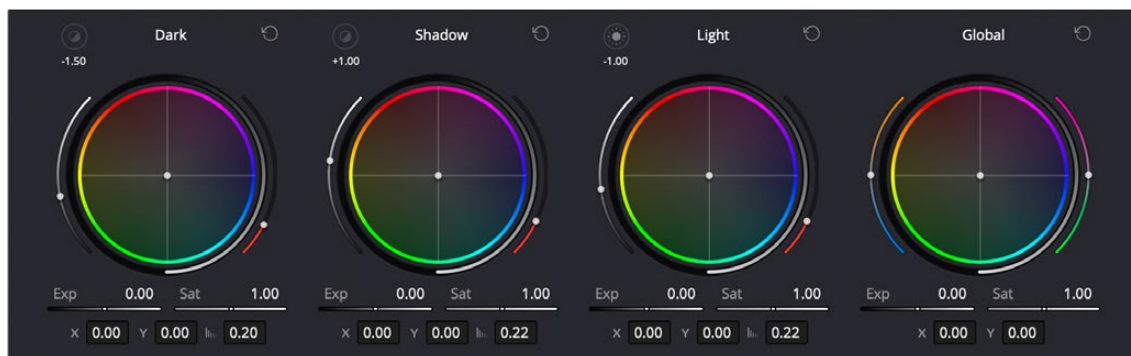
The HDR palette is specifically designed to enable fast and flexible primary grading of wide-latitude media for either SDR or HDR output. It's “color space aware” in that it works with Resolve Color Management to fit the known mapping of source image data within the HDR palette's own perceptually uniform operational color space. Using advanced color processing algorithms, this enables more efficient color and contrast adjustments, and perceptually uniform controls that make it easier to adjust all colors equally across the spectrum, while maintaining careful control of image saturation.



The HDR palette

A set of global controls, at the right of the palette, let you make foundational adjustments to the overall image, along with a set of Hue, Midtone Detail, Contrast/Pivot, and Black Offset controls at the bottom. Additionally, a set of overlapping zone-specific controls let you make color and contrast adjustments to specific ranges of image tonality. Altogether, the HDR palette lets you make primary grading adjustments that are both photographically naturalistic and as tonally broad or specific as you require, while minimizing unwanted artifacts.

Despite its sophistication, the HDR palette uses color balance and slider controls that should be familiar to any colorist or editor familiar with the more conventional Lift/Gamma/Gain controls. Whether you're a grading professional or just getting started with color, the HDR palette will enable great results with less hassle once you learn how to harness its power.



The controls panel of the HDR palette

TIP: Due to the increasing ubiquity of HDR-capable media and HDR streaming distribution workflows, the node HDR mode and HDR palette are both available without restriction in the free downloadable version of DaVinci Resolve.

What Makes the HDR Palette Special?

Before diving into the detailed use of the HDR palette's controls, it's important to understand a bit more about how the HDR palette processes image data, so you can better understand what advantages this tool presents.

Using the HDR Palette Without Color Management

While the HDR palette has been designed to work hand in hand with color management, you can use it in non-color managed workflows so long as you set it up correctly. The HDR palette option menu has Color Space and Gamma submenus that let you specify how you want to work. By default, both submenus are set to "Timeline" to reflect the color space you're choosing to work within in display-referred workflows.

Color Space Aware Controls When Using Color Management

The HDR palette really shines when you enable Resolve Color Management or ACES, because it's a color space aware palette that uses color management to full advantage. Being color space aware means that the color and contrast controls of the HDR palette conform themselves to the range of each clip's image data as mapped from the Input Color Space assigned to the source clip to the Timeline Color Space your program is working within. Practically, this provides two benefits:

- The controls of the HDR palette work and feel virtually identical no matter what type of source clip you're adjusting, and no matter which timeline color space you're choosing to work within.
- HDR palette adjustments made to one type of media will have a similar result when copied to other types of media. This makes matching shots and copying looks from one type of media to another easier than with previous tools.

NOTE: Because the HDR palette is color space aware, it's not necessary to enable the HDR setting on nodes for controls to work intuitively while working within a wide gamut Timeline Color Space, or delivering to an HDR format like ST.2084. The HDR palette takes care of this for you, automatically.

Perceptually Uniform Adjustments

Furthermore, the image data of each clip is converted from the Timeline Color Space to an operational color space used by the HDR palette, where the image is adjusted and then converted back to the Timeline Color Space again, ready for the next operation. The operational color space that's used by the HDR palette is perceptually uniform, which means that the range of values corresponding to each visible hue are distributed evenly throughout this color space.

While the HDR palette works well grading standard dynamic range (SDR) material, the underlying color science used by this palette also addresses issues that have inconvenienced colorists grading high dynamic range programs using traditional controls. These benefits include:

- Since the controls of this palette calculate all color transformations within a perceptually uniform color space, you will experience finer control over color adjustments in a more photographically intuitive way. For example, you may find that yellow hues are easier to grade within this palette, since there's a more even distribution of hues around the circumference of the color wheels.
- The HDR palette lets you make contrast adjustments without altering image saturation, meaning that unlike when you use the master wheels of the Lift, Gamma, and Gain panel, the HDR palette lets you raise image contrast without increasing image saturation, and lower image contrast without decreasing image saturation. You'll find this is particularly useful while boosting highlights

in HDR grades, since you can now do so without causing extreme saturation increases in your highlights. In fact, all adjustments you make to contrast with these controls will adjust the image while keeping saturation unerringly constant (that is to say, perceptually speaking; you will in fact notice small saturation changes in the Vectorscope). This, of course, excludes the saturation controls.

- The HDR palette's deep integration with Resolve Color Management also makes matching shots and copying looks from one type of media to another easier than with previous tools. Whether you're a professional, or just getting started with color, the HDR palette will enable better results with less hassle.

Superior Temp and Tint Controls

Because the HDR palette is color space aware, the Temperature and Tint adjustments in this palette are themselves made using the same kind of XYZ to LMS color space transform used by the Chromatic Adaptation Resolve FX plugin. The result is that Temp adjustments are photometrically accurate and create visual results similar to how the human vision system perceives changes to a scene due to a change in the color temperature of lighting.

Customizable Zones

Another unique aspect of the HDR palette is how it can be customized to suit a wide variety of working styles, specific tasks, and colorist preferences. Switching to the Zones panel reveals an interface for adjusting the Range and Falloff of each currently defined zone, using graphical controls overlaid onto a histogram of the current image, while a curve shows the user how the contrast and color adjustments they've made mathematically affect the RGB channels of the image within each zone. While the default zones preset provides an efficient starting point for many kinds of scenes, the varying tonal range of image data in different scenes, whether daytime, nighttime, interior, or exterior, will often require you to customize these zones to get the best results from adjustments using these controls.



The Zones panel lets you customize the tonal ranges that each set of zone controls affect.

Additionally, the sidebar at the left provides controls for adding and removing zones, so users can customize this palette to work exactly the way they want. Any customized preset can be saved for easy recall. This way, users have the freedom to work in whatever way suits them.

The Option menu provides fast access to additional ways of customizing this palette's operation, including the selection and management of presets, a choice of methods for numerically editing Color Balance adjustments, and a setting that determines whether or not the Global controls are always exposed, or should be banked to like the other zone controls.

The HDR Palette Interface

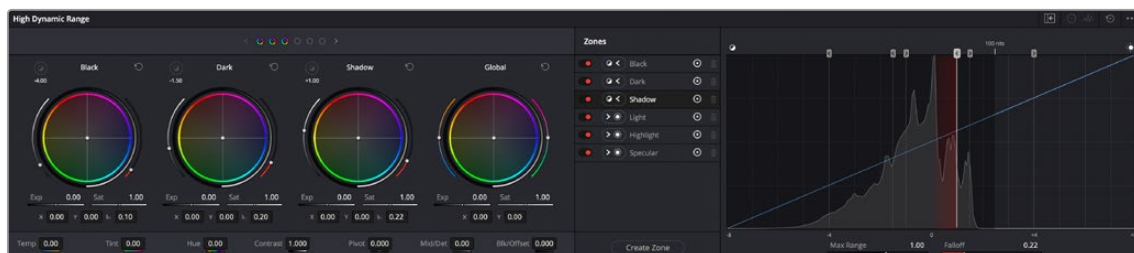
The HDR palette is divided into two panels. The first, containing Color Balance controls and custom sliders, is for making color and contrast adjustments. The second, containing the Zones sidebar and Zones graph, is for customizing the range and falloff of each of the zones that corresponds to each set of controls found on the controls panel. It's these zones that are used for making detailed image adjustments to tonally-specific areas of the picture.

By default, both panels appear within the same palette area, and you can switch between them using buttons at the top right of the palette. This is useful when you have limited screen real estate, or you're making simple adjustments. Alternately, you can click a button to expose both panels side-by-side, occupying both the left and middle palette areas of the Color page.



From left to right, the Side-by-side view button, the Controls panel, the Zones panel, the Reset control, and the Option menu

This Side-by-side mode makes it easy to see and customize how each zone is mapped to the source image data, while making adjustments at the same time.

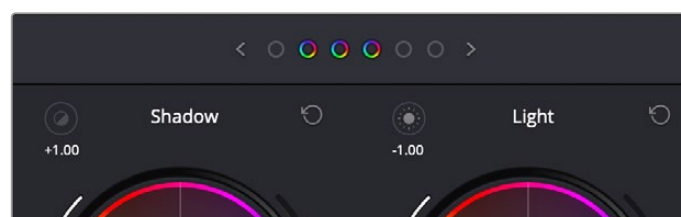


The HDR palette in Side-by-side mode; at left, the Color/Exposure/Saturation controls; at right, the Zone Customization controls

An Option menu at the far right of the title bar (shown as three dots) reveals the preset manager for this palette, as well as different GUI options for customizing how you want to use this palette.

Navigating Multiple Zones in the Controls Panel

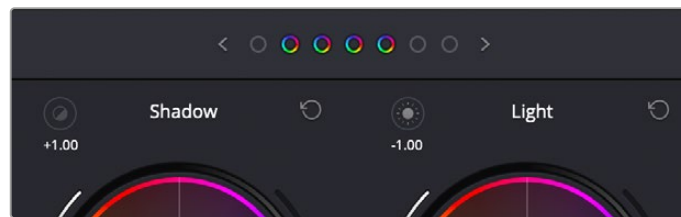
A row of buttons just under the HDR palette's title bar show you, of all available zones, which ones are currently adjustable using the visible controls below. While the default HDR palette preset has six zone-specific Zone controls, plus the Global controls, most users will only have room in the GUI for four sets of controls to be visible at any time. Meanwhile, grading control panel users will be constrained by how many trackballs their particular control panel has.



The Zone bank controls

Since you'll almost always have more zones available than you have room to adjust them, this interface lets you see which zones you're currently using, which zones are mapped to a control panel, and which other zones are available for you to switch to. By default, there are three active zones shown at the top, corresponding to the three sets of Zone controls to the left of the Global controls. Zones that are available, but hidden, are shown in gray to the left and/or right of the currently visible zones, which appear in color. This way, you can bank back and forth among three zones at a time, while always having the Global controls available for adjustment.

There's also a setting in the HDR palette's Option menu, "Bank Global With Color Wheels," that lets you use all four GUI control clusters in this palette to show Zones controls, making it faster to jump among tonally-specific adjustments. In this mode, there are four active zones shown at the top, corresponding to the four sets of Zone controls being shown. However, in this mode you must page all the way to the right to expose the Global controls, which are banked along with all the other zones.



The Zone bank controls set to show four zones when the Global controls are banked with all the other zones

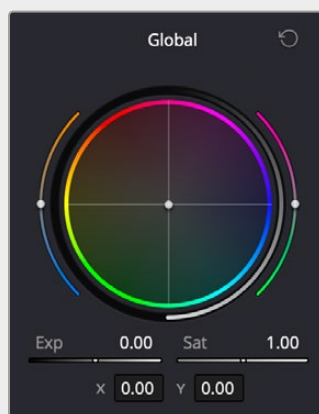
Using these controls, you can switch your visible control among all available zones similarly to how audio mixers switch among banks of tracks with a fader panel.

Methods of switching among different Zone controls:

- Click any gray zone button to the left of the currently visible zones to move all zones so that the leftmost visible zone appears at that one.
- Click any gray zone button to the right of the currently visible zones to move all zones so that the rightmost visible zone appears at that one.
- Click the Left or Right Arrow buttons to either side of the zones indicators to move all zones one to the left or one to the right.

Making Global Adjustments

Generally speaking, if you're using the HDR palette to do primary grading, you'll probably want to begin with the Global controls. By default, these are found at the right of the Controls panel.



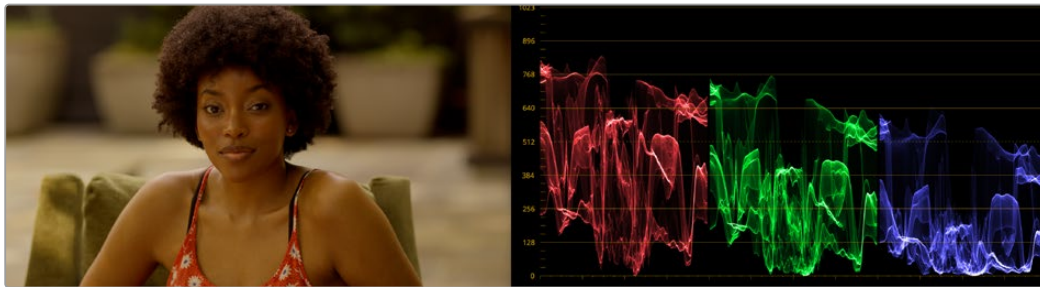
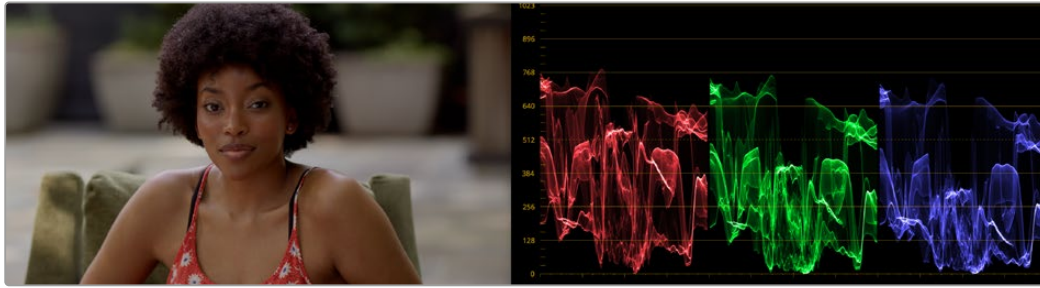
The graphical Global control cluster at the right of the HDR palette

These controls let you take the initial step of making an overall adjustment to the image. To adjust color, you have a color balance wheel, temp and tint sliders (to the left and right of the wheel, and also numerically below), and saturation adjustments. To adjust image contrast, a Global Exposure control lets you stretch overall image contrast up to brighten the image, or compress it down to darken it, relative to the current Black Offset level, which pins pixels corresponding to minimum black in place. A pair of Contrast/Pivot controls at the bottom of this palette let you manipulate overall color contrast by stretching highlights and shadows out or in relative to the pivot point, expressed in stops.

While Global color adjustments are somewhat similar to the offset, temp, and tint controls found in the Color Wheels or Log Wheels controls, there are important differences, described in detail below. The Global Exposure control, on the other hand, operates very differently from Master Offset.

Detailed Explanation of Global controls:

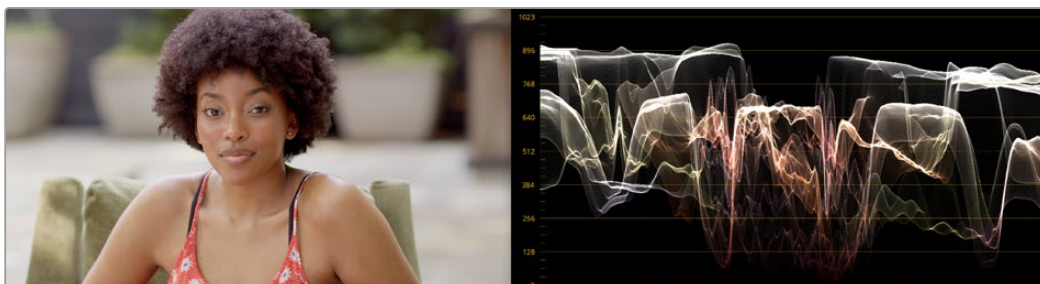
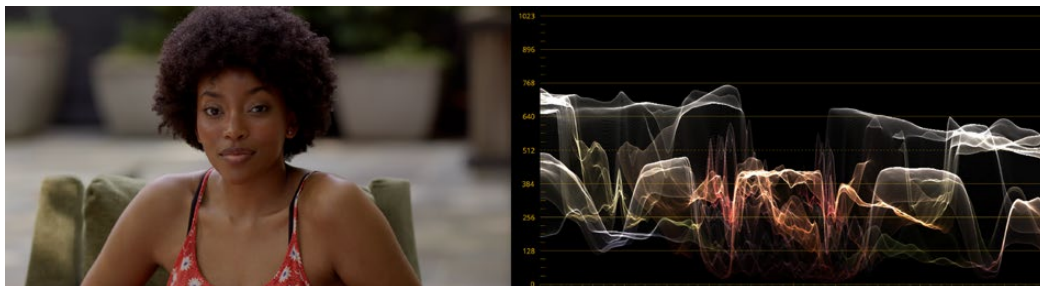
- **Global Color Balance:** A color balance control lets you simultaneously adjust the red, green, and blue channels, somewhat linearly throughout the tonal range of the image, to freely rebalance the color. However, RCMv2 automatically maintains a smooth saturation and exposure rolloff in the maximum highlights and minimum shadows, so this effect becomes less pronounced in the blackest blacks and the whitest whites of the image (this appears more pronounced in test gradients than real world images). While adjustments made with this control won't lower the value of the darkest pixels in the image below the Black Offset level, individual color channels may dip below this level in order to preserve colorfulness in darker shadows. The resulting color adjustments made with this control are similar to how optical color filters placed over a lens would affect an image.



(Top) The original image, (Bottom) A Global Color Balance adjustment affecting the overall image, with a rolloff seen in the shadows and highlights of the superimposed ramp gradient

- **Global Exposure:** Brightens or darkens the entire image. Dragging to the right stretches image contrast by raising the highlights while keeping the shadows pinned relative to the Black Offset value. Dragging to the left lowers image contrast by lowering the highlights while keeping shadows pinned relative to the Black Offset value. For more explanation about the function of Black Offset, see the following section.

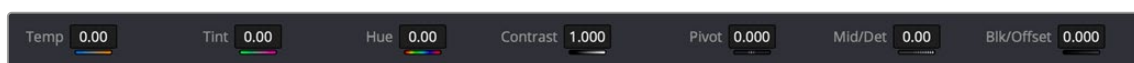
Keep in mind that all contrast adjustments in the HDR palette keep saturation constant as contrast increases or decreases. This means that, when making large contrast adjustments to HDR images, image saturation will remain perceptually the same.



(Top) The original image, (Bottom) Raising Global Exposure

- **Global Saturation:** Raises or lowers the intensity of color throughout the image. Since contrast adjustments don't affect saturation, you must exercise manual control over any desired changes to image saturation.
- **Global Color Balance Value Fields:** Two numeric values represent how the color balance controls center handle position is represented. There are two ways of numerically showing this: as X and Y, or as Angle and Strength. You can choose either method from the HDR palette's Option menu; whichever method you choose determines how the center handle of all color balance controls is manipulated if you adjust these values directly.

At the bottom of the HDR palette, additional Global Temp and Tint, Hue, Midtone Detail, and Contrast/Pivot controls let you make other kinds of global adjustments to image color, detail, and contrast, in ways that are very similar to the identically named controls in the Color Wheels or Log Wheels controls. Additionally, there's a Black Offset control that lets you set the minimum allowable black level in the current shot.



Global controls found at the bottom of the HDR palette

Detailed explanation of additional Global controls:

- **Temp:** This global temp adjustment lets you warm or cool the image with a range from -4000 to +4000 based on a kelvin scale. Adjustment follows the spectral locus line in the CIE graph: the intensity of adjustment is naturalistic. To minimize visual artifacts, this control is capable of warming the image more aggressively than cooling it.
- **Tint:** This global tint adjustment lets you add “plus green” or “minus green” adjustments with a range from -100 to +100 on an arbitrary scale.
- **Hue:** Rotates all hues of the image around the full perimeter of the color wheel. The default setting of 50 shows the original distribution of hues. Raising or lowering this value rotates all hues forward or backward along the hue distribution as seen on a color wheel.
- **Midtone Detail:** When this parameter is raised, the contrast of regions of the image with high edge detail is raised to increase the perception of image sharpness. The effect is sometimes referred to as definition. When this parameter is lowered to a negative value, regions of the image with low amounts of detail are softened, while areas of high detail are left alone. 0 is unity. The range is -100 through +100.
- **Contrast:** This one parameter lets you increase or reduce the distance between the darkest and lightest values of an image, raising or lowering image contrast. The effect is similar to using the Lift and Gain master controls to make simultaneous opposing adjustments. Bright and dark parts of the image are pushed apart or brought together about a center point defined by the Pivot parameter. Unlike the Contrast control in the Primaries palette, the Contrast control in the HDR palette ignores the “Use S-curve for contrast” setting in the General Options panel of the Project Settings.

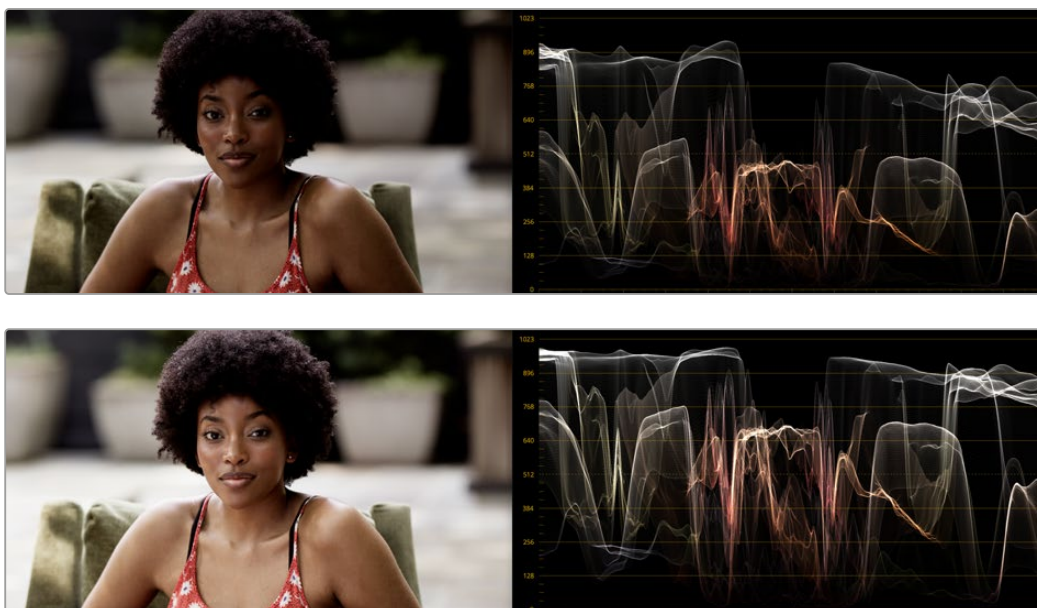
Keep in mind that all contrast adjustments in the HDR palette keep saturation constant as contrast increases or decreases. This means that, even when making large contrast adjustments to HDR images, image saturation will remain perceptually the same to your eye, with small changes made to the actual saturation of the image in order to maintain this perception. Be aware that the tone mapping settings in RCMv2 will have an effect on how much saturation is maintained in the top highlights and bottom shadows of the image, versus being automatically rolled off at the outer boundaries of the signal.



(Left) The original image, (Right) Raising Contrast, saturation is kept perceptually constant, except at the outer boundaries of the signal in the top highlights and bottom shadows (depending on how you have your tone mapping set in RCM).

- **Pivot:** Changes the center of tonality about which dark and bright parts of the image are stretched or narrowed during a contrast adjustment. This lets you change how contrast is distributed among the shadows and highlights. The default value of 0 pushes highlights above 50% gray higher, and shadows below 50% gray lower, in equal proportion to one another. Changing the pivot point changes the percentage of image tonality about which contrast is either expanded or compressed.

Darker images may require a lower Pivot value to push more levels up and avoid crushing the shadows too much when stretching image contrast, while lighter images may benefit from a higher Pivot value to push levels down and increase shadow density.



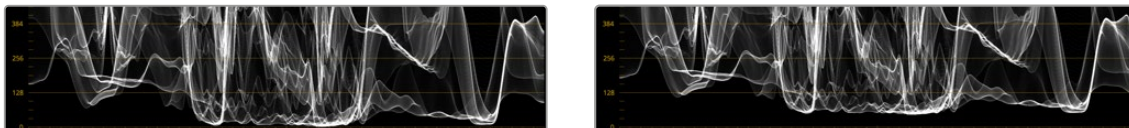
(Top) Altering contrast by raising Pivot, which redistributes most of the contrast expansion down towards shadows, (Bottom) Altering contrast by lowering Pivot, which redistributes most of the contrast expansion up towards the highlights

- **Black Offset:** Defines the darkest allowable part of the image. No adjustment by any other control in the HDR palette will ever go below this level, except for the occasional color channel if the shadows become extremely saturated. For more detailed information on this extremely important control, see the very next section.

Black Offset Explained

Of all the Global controls, the Black Offset control is one of the most deceptively important. Black Offset lets you define the darkest pixels of the image. Adjustments made to Black Offset do not offset the entire signal, they only affect the very darkest portion of the signal, letting you raise or lower the black point of the image while smoothly blending the resulting adjustment into the unadjusted portion of the image. The result is that you can compress or expand the bottom of your signal.

If you raise Black Offset, you can add “flaring” to the image, similarly to how light flaring within a lens might lighten the darkest part of the image.



(Left) The original image, (Right) Raising Black Offset in order to add flaring to the darkest pixels of the image

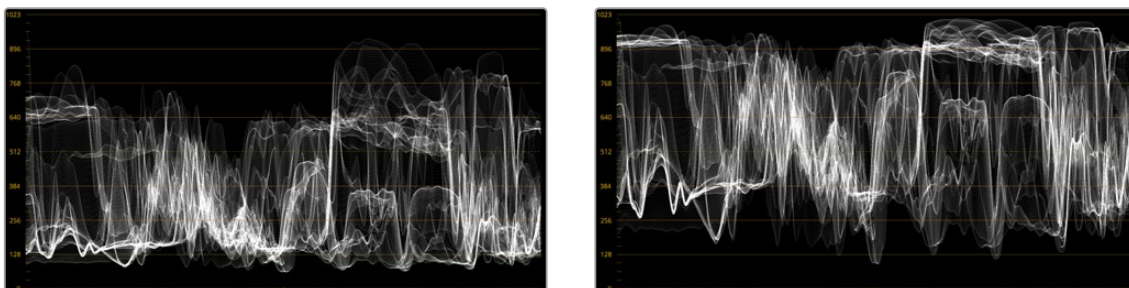
Alternately, you can lower Black Offset to lower the very darkest pixels in the image. If the source image is dark enough, this control can lower the darkest pixels below 0, however this image detail is preserved in the color image processing pipeline

The most important thing to understand about this control is that level to which you set Black Offset becomes the new level at which Global exposure adjustments are made. In the following example, the first image shows the original color of a clip that was exposed to be dark. In the second image, Black Offset is adjusted to raise the darkest parts of the image, compressing them relative to the midtones and introducing a pleasing flaring effect to the shadows.



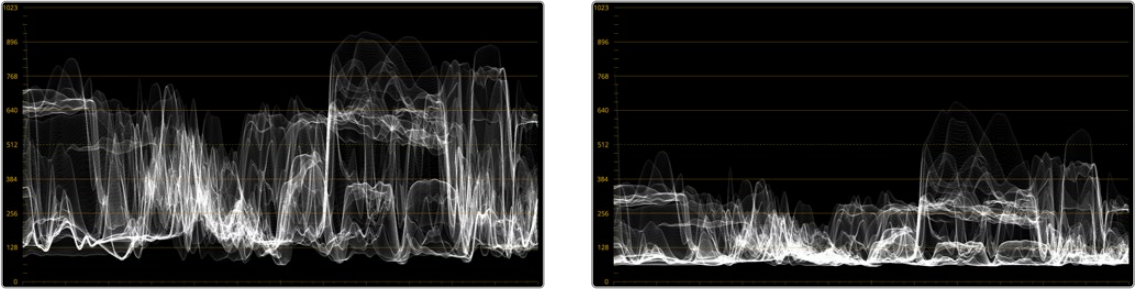
(Left) The original image; (Right) The image with Black Offset raised

In the following image, Global Exposure is raised, and you can see in the Waveform scope that the exposure change stretches the contrast of the image up, with the darkest pixels “locked” to the Black Offset you set.



(Left) Before, (Right) After raising Global Exposure to stretch image contrast out from the Black Offset level; notice that the bottom of the signal continues “touching” the same level after global exposure is raised.

The Black Offset level you set also becomes the minimum picture level when you lower exposure, either using the Global Exposure changes, or when you use any of the “darkening” zone controls such as Shadow, Dark, and Black. The following image shows the result of lowering Global Exposure; the shadows of the image are compressed more heavily than the highlights as exposure is lowered, with the result that the minimum value of the image remains at the Black Offset level you set. With shadow detail impressively preserved, this compression rolls off smoothly and nonlinearly through the midtones so that the image continues to look as natural as possible.



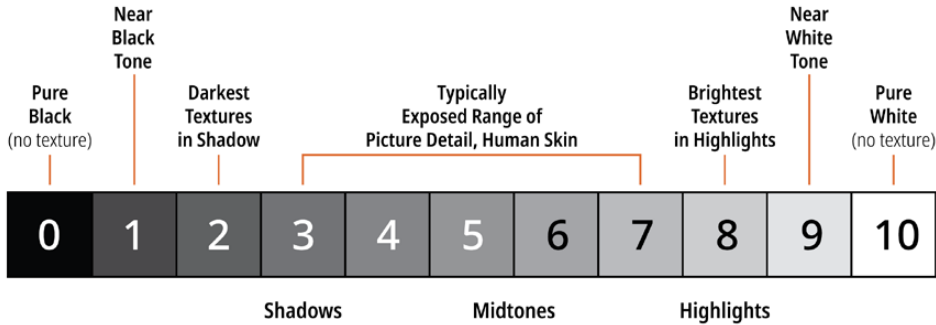
(Left) Before, (Right) After lowering Global Exposure to compress image contrast down to the Black Offset level; notice the signal never goes lower than the Black Offset level

This is also true when using other darkening zone exposure controls, such as the Shadow control, which selectively pushes the exposure of the darker part of the image down, while leaving the highlights alone. While reducing Shadow exposure, the darkest pixels still compress so that nothing goes beneath the Black Offset level you set.

Bottom line, you can adjust Black Offset at any time, either before or after any other Global or Zone adjustments, to fine tune the image as necessary.

Making Zone-Based Adjustments

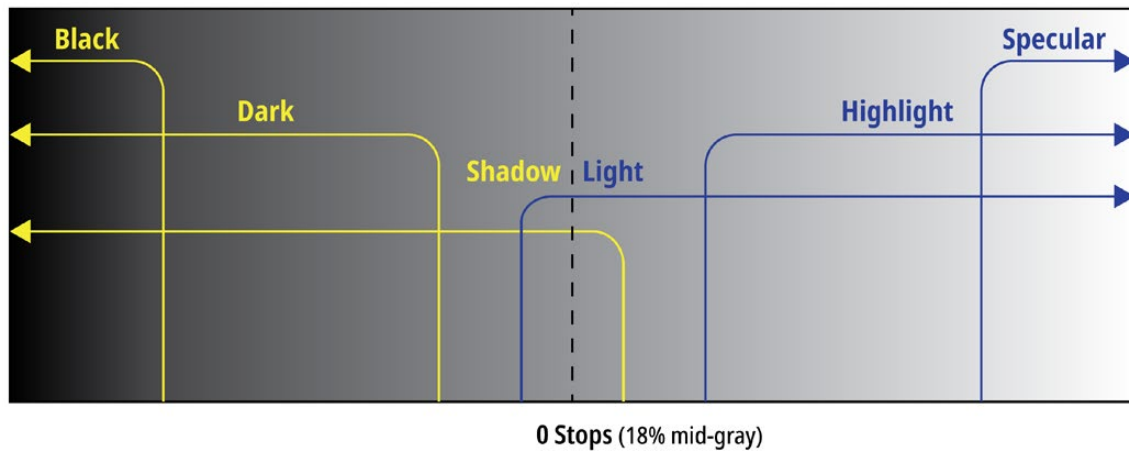
After you’ve made whatever initial global adjustments you want, you can optionally make more tonally-specific adjustments to the image via the Zone-based color and contrast controls to the left. These divide the image into multiple overlapping “zones,” which are somewhat similar in principal to Ansel Adams’ zone system, which divides images into tonally-specific regions based on image luminance, from pure black, through progressively lighter shadows, to 18% gray as the middle value centering the midtones of most images, and then up through progressively lighter highlights on the way to the last zone of pure white.



Simplified illustration of Ansel Adams’ zone system

While Adams’ zone system was meant to teach people how to think about using the available range of the photographic medium when exposing images, the zones of the HDR palette let you put this concept into practice by allowing you to make tonally-specific Color Balance, Saturation, and Exposure adjustments that only affect the range of shadows or highlights that fall within that zone.

Using the default preset, the shadows of your image are divided into overlapping Shadow, Dark, and Black zones, while the highlights of your image are divided into separate overlapping zones for Light, Highlight, and Specular ranges. The chart below illustrates the relationship of these zones to one another.



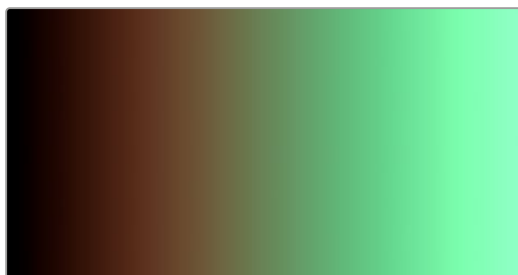
Approximation of the overlapping zones in the HDR palette that can be individually adjusted

NOTE: All of the examples in this section have Timeline to Output Tone Mapping set to None, to keep the ramp gradient linear for purposes of this explanation.

Zone-Based Color Adjustments

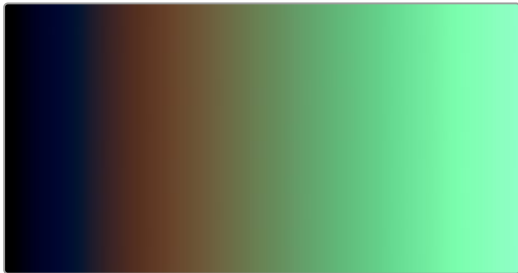
Understanding how adjustments made using these overlapping zones interact with one another is key to knowing what you can do using the HDR palette. Keep in mind that every zone is customizable, but here's how the default zone definitions work to let you make detailed image adjustments. For clarity, Zone adjustments are shown made to a linear ramp gradient image, to demonstrate exactly what tonal portion of each image is affected by each control.

- The Light and Shadow zones are the most broadly defined zones. Together, they cover the entire tonal range of your image. They softly overlap one another by two stops at the center of the tonal range (50% gray or 18% exposure); this is where adjustments made to Light mix together with adjustments made to Shadow, resulting in both adjustments softly blending together. Making exposure adjustments to these two zones lets you manipulate overall image contrast by selectively compressing/expanding contrast in the shadows and/or highlights relative to one another. Using the color controls lets you make broad color adjustments to the upper (highlights) and/or lower (shadows) range of image tonality.



Color adjustments to a ramp gradient show the Light (tinted green) and Shadow (tinted orange) zones

- The Dark and Black zones overlap two progressively lower luminance ranges of the Shadow zone. Dark is useful for adjusting the color or contrast of the deeper shadows, while Black is useful as a trim that lets you manipulate the very darkest parts of an image. All adjustments you make to Dark are mixed with any adjustments made to Shadow, and all adjustments made to Black are mixed with the adjustments made to Dark and Shadow. In this way, all color adjustments you make to overlapping Shadow zones mix smoothly together to produce the final result, as seen in the following ramp gradient.



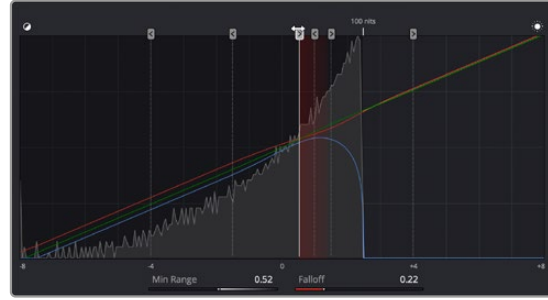
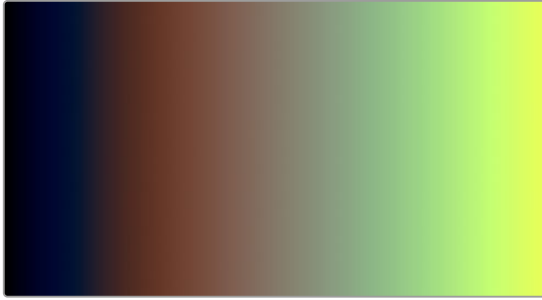
Color adjustments to a ramp gradient show the Dark (tinted blue) overlapping the Shadow (tinted orange) zone

- The Highlight and Specular zones overlap two progressively brighter luminance ranges of the Light zone. Highlight lets you adjust the color and/or contrast of the brightest highlights, while Specular is useful as a trim that lets you manipulate the very brightest highlights of an image. All adjustments you make to Highlight are mixed with any adjustments made to Light, and all adjustments made to Specular are mixed with the adjustments made to Highlight and Light. In this way, all color adjustments you make to overlapping Highlight zones mix smoothly together to produce the final result, as seen in the following ramp gradient.



Color adjustments to a ramp gradient show the Highlight (tinted yellow) overlapping the Light (tinted green) zone.

As you can see, each one of these zones allow the specific adjustment of whichever parts of the image fall within the luminance range of that zone (as defined on the Zones panel), while providing control over how smoothly one adjustment fades into the next, to prevent unwanted contouring. As you make adjustments to the Range and Falloff of each zone using the different sets of zone controls, they work together to smoothly manipulate the contrast and color of the image, similarly to if you were using a more precise and constrained version of the Custom Curves. In fact, you can see the actual adjustment you're making by exposing the Zone graph.



(Left) Raising the Range of the Light zone to expand the Shadow zone's range of influence, (Right) The same adjustment seen in the Zone graph

Working this way allows a tremendous specificity of adjustment, without the need to use curves or qualifiers. For example, zone-specific Color Balance controls allow you to grade the highlights and shadows separately using the Shadow and Light zone controls to quickly create a warm/cool split lighting effect. This makes the creation of dynamic grades more efficient and more creative.



(Left) The original image, (Right) Using the HDR palette to warm the highlights and cool the shadows (exaggerated for print)

In this next example, the Dark zone's Color Balance control is used to add a splash of green to the darker shadows of the image, while the Black control is used to neutralize this green in the very blackest shadows.



(Left) The original image, (Right) Using the HDR palette to add green to a narrow range of shadows

Zone-Based Exposure Adjustments

Each zone has its own Exposure control that allows you to stretch image contrast from that zone's Min or Max range boundary either down towards the shadows, or up towards the highlights. As seen on the Zone graph (found on the Zone panel), exposure adjustments are made on a scale measured in stops, to provide a more photographic experience when grading wide dynamic range material.



The scale of the Zone graph shows you the shadow and highlight ranges, in stops, relative to 0 representing 18 percent photographic gray

On this scale, 0 stops corresponds to photographic 18% gray, at the center of each image's tonal range. Shadows then fall to the left over -8 stops, while highlights fall to the right over +8 stops, where each stop represents either twice the amount of light (stopping up) or half the amount of light (stopping down).

In conjunction with Resolve Color Management, this range is designed to accommodate any SDR or HDR range you'll be mastering to. All HDR palette controls are designed to accommodate up to 16 stops of exposure. By comparison, most modern digital cinema cameras claim to be able to capture somewhere between 13-19 stops of dynamic range, while a white paper from the ASC states that modern film negatives are capable of capturing 14 stops of dynamic range.

Unlike the Lift/Gamma/Gain or Log master level controls, the Exposure adjustment of each zone starts at a specific boundary of image tonality, and stretches all the way out through the highlights or shadows, through the maximum or minimum signal level allowable. Since Shadow and Light are the two zones that are centered on the midtones of the image, they're the best zones for seeing how this works. Because of this, these are also probably the first two zones you'll want to adjust if you're making more specific exposure changes to the image.

Using the default preset, Shadow exposure adjustments affects everything in the image from stop 1 down through absolute black.



(Left) The original image, (Right) Darkening all shadows by lowering the Shadow exposure adjustment

Meanwhile, Light exposure adjustments affect everything in the image from -1 stops all the way through maximum white.



(Left) The original image, (Right) Lightening all highlights by raising the Light exposure adjustment

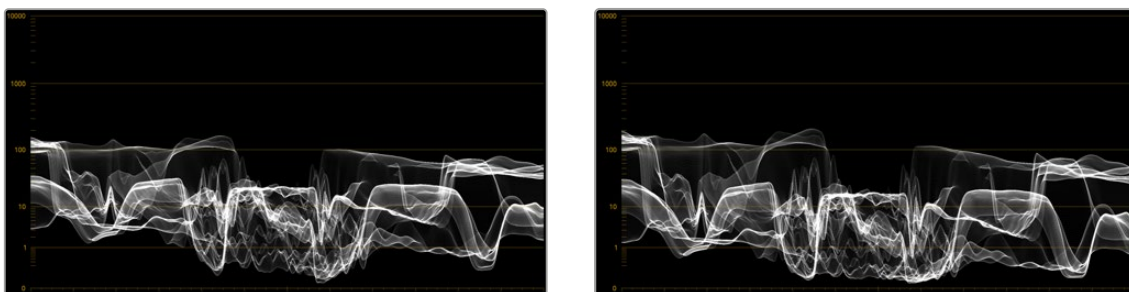
This ability to make open-ended adjustments to image contrast in different overlapping ranges of shadow and highlight tonality makes these controls uniquely suitable for HDR grading, while also being completely suitable when grading at SDR levels. Also, keep in mind that all contrast adjustments in the HDR palette keep saturation constant as contrast increases or decreases.

This is particularly useful when making large contrast adjustments to HDR images, for which a 300 nit expansion of highlight contrast using the Lift/Gamma/Gain or Log controls would create a huge saturation increase. Using the HDR palette to expand highlight contrast keeps image saturation perceptually the same, while offering zone-specific saturation controls that let you choose whether to increase or decrease the saturation falling within that zone.

Exposure adjustments to zones that overlap other zones are smoothly combined to allow targeted contrast adjustments, while making it relatively easy to avoid unwanted artifacts, like contouring or solarizing due to overzealous inversion of the curves caused by these operations.

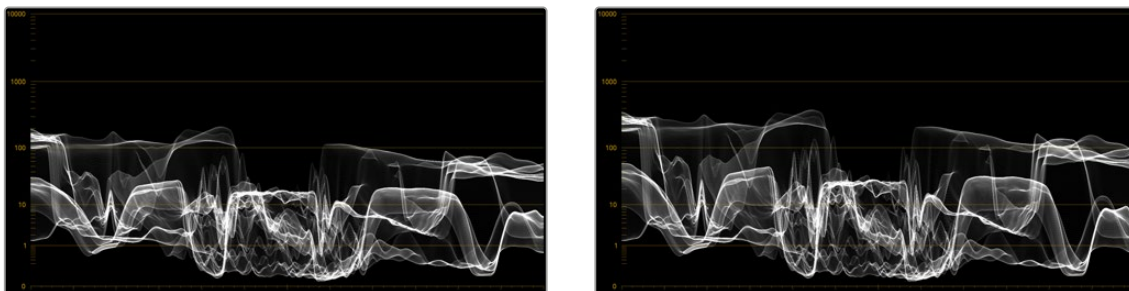
An Example of Zone-Based Exposure Adjustment

To see this in action, let's see what happens when we make some overlapping adjustments to the Light, Highlight, and Specular zones in order to grade some HDR highlights, focusing our attention on the Waveform scope, since it's difficult to see HDR grades in print. Keep in mind that you won't usually be making Zone-based Exposure adjustments by themselves. This example image has had a Contrast/Pivot adjustment to push the midtones and shadows down, to give the shadows greater definition. Then, Black Offset is raised to lift the darkest shadows up just a little bit, so they can breathe. This leaves us with an image that's mostly under 200 nits to begin with. A perfect starting point for us to begin sculpting those highlights.



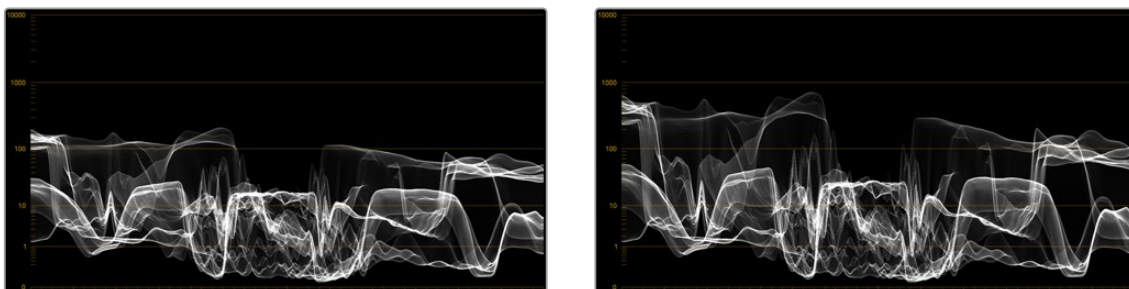
(Left) The original image; (Right) After changing Contrast/Pivot and raising Black Offset

Using the default preset, making an exposure adjustment to the Light zone smoothly raises everything in the source image above -1 stops on the Zone graph. Practically, it's stretching up all levels above 10 nits on the HDR scope. This smoothly boosts the brighter part of the image, while leaving the shadows down where we put them.



(Left) The previous adjustment; (Right) After raising Light Exposure

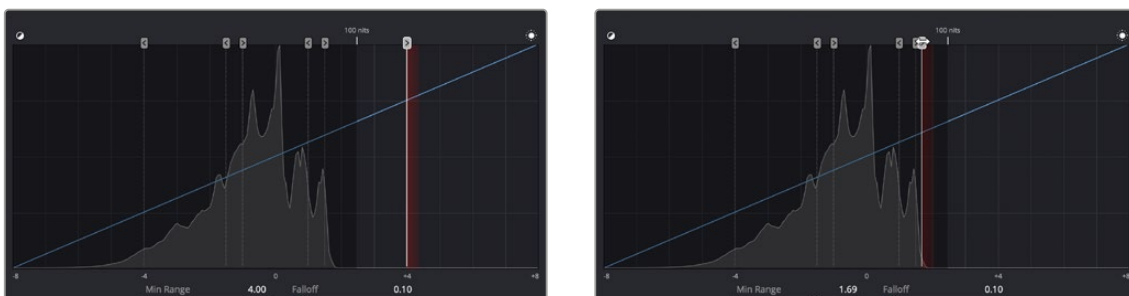
At this point, making an exposure adjustment to the Highlight zone boosts all levels above 1.5 stops on the Zone graph. Again, practically speaking, all levels above 100 nits are being boosted, brightening the top highlights. In the image, you can see that we're creating a greater differentiation between dimmer and brighter highlights in the image, creating highlight contrast with greater detail where once there were simply flat pools of light.



(Left) The previous adjustment; (Right) After raising Highlight Exposure

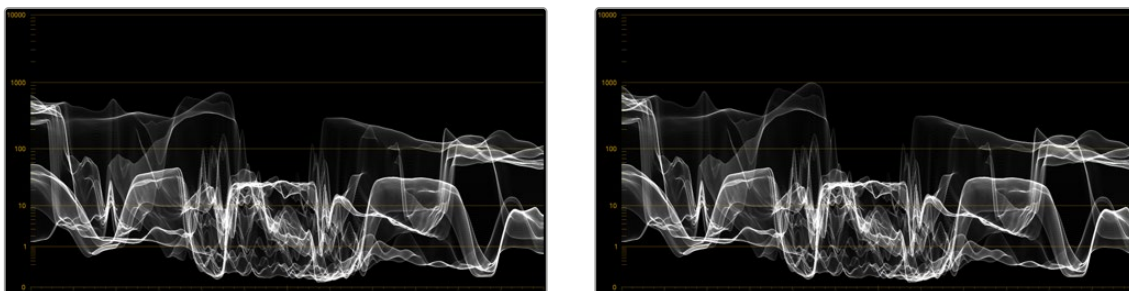
Now, this might be a fine place to stop, but since we're interested in turning the HDR in this image up to 11, we can put the highlights over the top by making one more Exposure adjustment, to the Specular zone. The Specular zone starts at 4 stops and is designed to let you boost or attenuate only the very brightest, hardest highlights, likely corresponding to specular details such as chrome reflections, eye glints, and directly photographed light sources. You won't always use this control, but it's useful to know about.

In this example, the default Specular Zone Min Range control is outside of the available image data in the image (seen as the right edge of the histogram in the Zone graph), so making adjustments with this control won't do anything. However, dragging the Min Range control to intersect the end of the histogram will allow these controls to affect the very brightest pixels in the image.



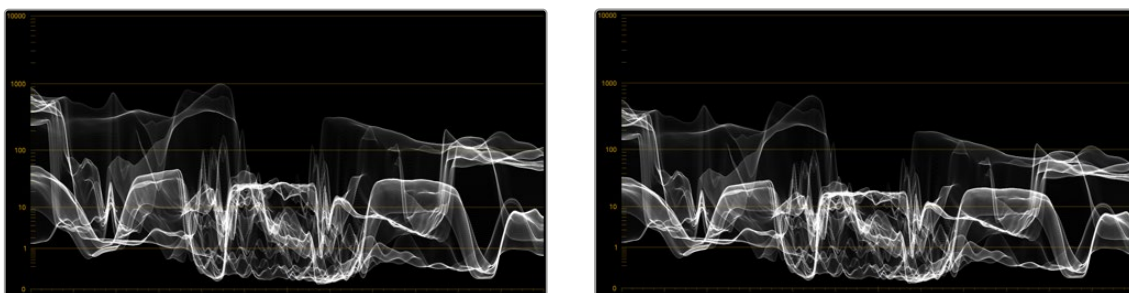
(Left) The original distribution of zones in the Zone Graph; (Right) Dragging the Specular Min Range handle to intersect the brightest highlights in the image histogram, to allow the Specular controls to affect that part of the image

Now, the Specular Exposure control lets us push the very brightest pixels of our image up, further differentiating them from the other highlights, and widening the audience's sensation of highlight contrast.



(Left) The previous adjustment; (Right) After raising Specular Exposure to bring the very brightest highlights to 1000 nits

Now, having selectively boosted the Highlight and Specular zones, you might decide that you like the differentiation that's been created among different brightnesses of highlights, but the overall highlights are too bright. This is easily fixed by lowering the Light zone's Exposure adjustment. As a result of this one adjustment, all of the highlights are scaled down, while the adjustments you've made to the Highlight and Specular zones maintain a relative influence on the parts of the image they're brightening. The Shadow, Dark, and Black controls work similarly, but for the shadows.

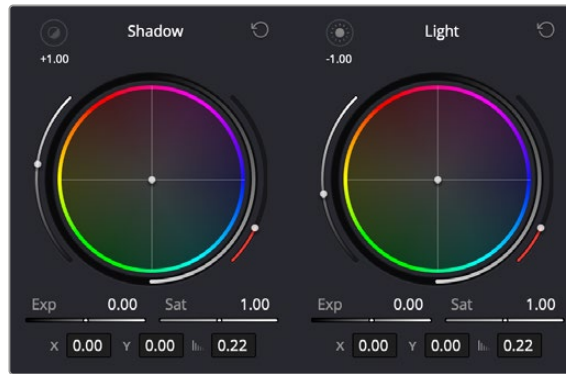


(Left) The previous adjustment; (Right) After lowering Light Exposure, all levels smoothly scale down to reduce the brightness of the highlights while keeping the differentiation you've created with the Highlight and Specular controls.

Now that you've seen how all overlapping Zone adjustments work together to create a seamless adjustment, it's time to look at the Zone controls in more detail.

Zone Controls

Each individual zone also has controls for Color Balance and Saturation, which let you make focused adjustments that fall within specific ranges of image tonality, all without needing to use a qualifier. Using all of these zones together, you can make fast, precise, and smooth adjustments to the image that feel incredibly natural. Despite the name of the palette, this allows careful adjustment of SDR images, as well as creative adjustment of the spectacular range of highlights that HDR enables. There are also Range and Falloff controls to the left and right of each cluster of controls that let you redefine each zone's area of influence, even if the Zone graph is hidden.

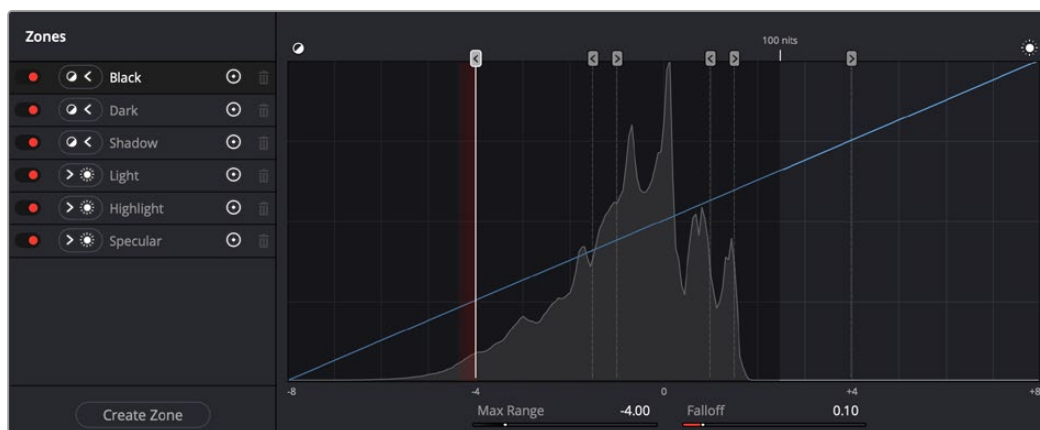


The Shadow and Light zone controls next to one another; you can see the Min and Max indicators, along with the Range value (in stops) at the upper left of each group of controls.

- **Range indicator:** An icon and value at the upper-right-hand corner of each zone control cluster shows that zone's range. A numeric value shows the range value, which is the Luminance level of image tonality at which that zone's adjustment begins. This value is expressed in stops, while an icon shows you whether it's a maximum range (affecting shadows) or a minimum range (affecting highlights).
Clicking the icon shows a temporary preview in the Viewer of which parts of the image will be affected by the controls of that zone. Affected areas appear in full color, while unaffected areas appear black.
- **Zone-specific Color Balance:** A color balance control lets you readjust the relative strength of the red, green, and blue channels within the current zone of image tonality.
- **Zone-specific Exposure:** Zone-specific Exposure controls work in conjunction with an adjustable pivot that defines the Luminance level that each exposure adjustment starts from. To accommodate both shadow and highlight adjustments, there are two kinds of pivots, labeled Maximum and Minimum range, and indicated using an icon. Max Range controls, such as the Shadow control, start at a maximum value with exposure adjustments stretching image contrast downward towards the shadows. Min Range controls, such as Light, start at a minimum value with exposure adjustments stretching contrast upwards towards the highlights. Each zone also has a falloff that specifies how much of that zone overlaps neighboring zones (starting at the pivot level) in order to blend overlapping adjustments softly enough to prevent contouring.
- **Zone-specific Saturation:** Adjusts the intensity of color within the current zone of image tonality.
- **Min/Max Range:** Defines the level of image tonality at which that zone's adjustment starts. This value is expressed in stops. Identical to the Min or Max Range slider in the Zone panel. Useful for tweaking Range while the Zones graph is hidden.
- **Zone Falloff:** Defines how softly this zone fades into all overlapping zones. Identical to the Falloff slider in the Zone panel, the value is visible in the virtual slider below. Useful for tweaking Falloff while the Zones graph is hidden.
- **Zone-specific Color Balance Values:** Two values expose the current color balance operation numerically. The option menu lets you choose between representing these as X and Y, or as Angle and Strength.

Customizing Zones Using the Zone Panel

Similarly to the controls of the Log palette, how these zones overlap can be adjusted in order to make each zone's controls work best for the tonal distribution of the images in any given scene. Zone overlap is governed by individual pairs of Range and Falloff controls, so you can make precise adjustments no matter what the distribution of tonality is in the image you're grading. All zones are organized and customized using the Zone panel, which is divided into the Zones sidebar to the left, and the Zone graph to the right.



The Zone panel of the HDR palette

The Zone Sidebar

The Zone sidebar shows a list of all the zones in the current preset. This list is sorted by range values, with minimum range zones (affecting shadows) appearing at top and maximum range zones (affecting highlights) appearing below. Clicking any zone in the list selects that zone, which also selects that zone's boundary handle in the Zone graph at the right.

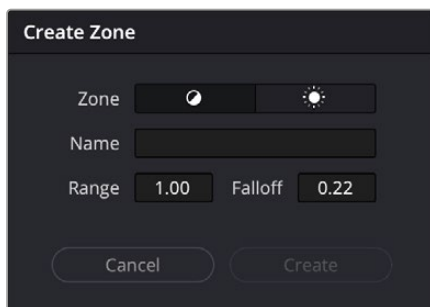
Zone sidebar controls:

- **Enable Zone toggle:** Toggles let you turn each zone on and off. Disabled zones will not be rendered.
- **Max/Min button:** Defines whether the adjustment of a particular zone affects the image from the current range value and below (Max) or the current range value and higher (Min). This setting is typically used when creating your own custom preset, to define which zones stretch down toward the shadows and which zones stretch up toward the highlights.
- **Show/Hide Zone button:** This button lets you hide sets of zone controls you don't want to adjust any further, leaving more room in the UI and on your control surface for the zones you do want to adjust. For example, if you decide you don't need to use the default Black and Specular zones, you can hide them so that only the Dark, Shadow, Light, and Highlight zones are available to switch among. Zone controls with adjustments made to them continue to affect the image even if the controls are hidden.
- **Delete Zone button:** Deletes that zone. This is primarily useful when you're creating your own zone preset. At the time of this writing, you cannot delete any of the default zones found in the default Zone preset.

Additionally, you have the option of adding zones, and saving zone presets to create your own custom ways of adjusting the image.

To create a new zone:

- 1 Open the Zones panel.
- 2 Click Create Zone, at the bottom of the Zones sidebar.
- 3 When the Create Zone dialog appears, choose the Zone type you want that zone to have (Dark/Min Zone or Light/Max Zone), enter a Name, and define the Range and Falloff values. Range, Falloff, and Zone type are editable later, so don't worry if you don't set them up perfectly the first time.



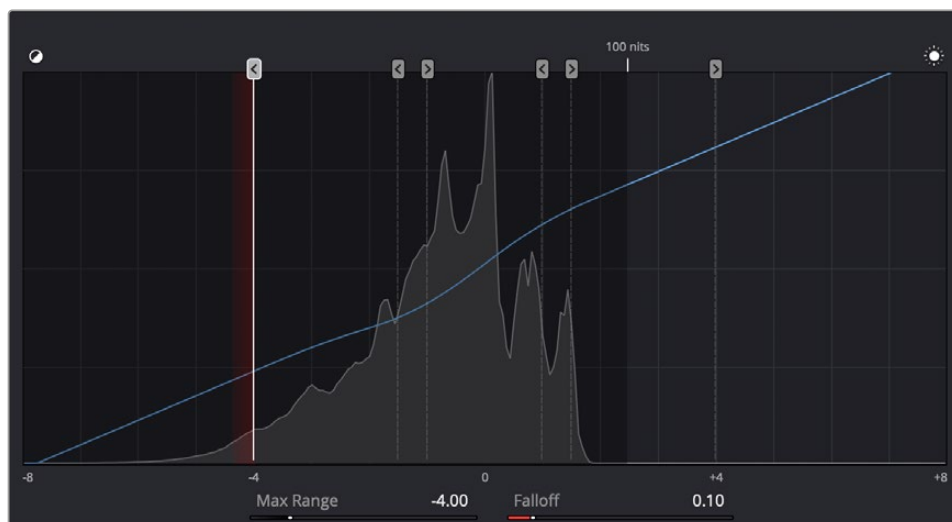
The Create Zone dialog

- 4 Click Create. That zone appears added to the list, which is sorted in the order in which Range is defined, with Dark Zones appearing at the top, and Light Zones appearing at the bottom.

Zones Graph

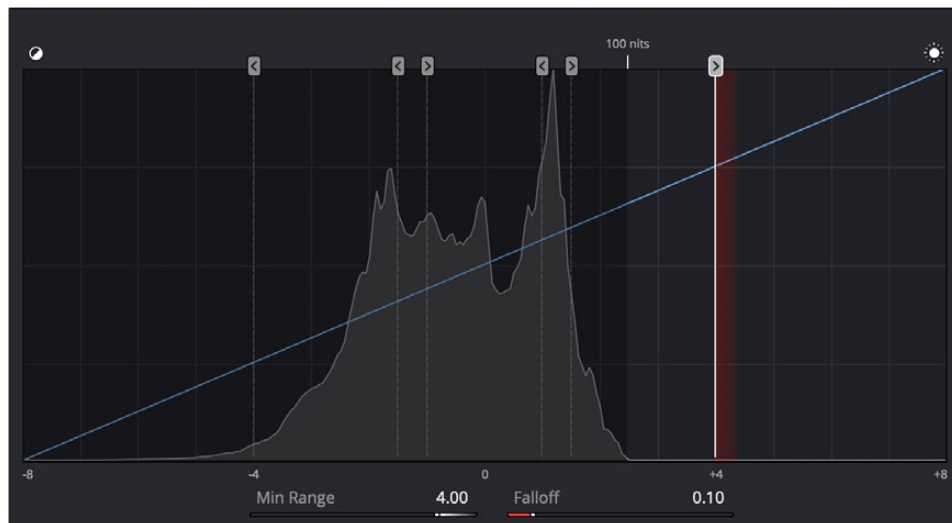
The Zone graph is a key component of the HDR palette's use. Along the bottom, a scale in stops shows the range over which the HDR palette operates. At the top, handles let you adjust the editable range boundaries that define the tonal range of each zone's operation, superimposed over a histogram showing the current image as input into the current node. Running through the middle of the Zone graph, a curve shows you all the Global and Tonal color and contrast adjustments being made by the HDR palette in the current grade.

The relationship of the range boundaries with the source image histogram in this graph is extremely important. The range boundary for a particular zone must intersect with the part of the source histogram you want to adjust for the controls of that zone to have any effect. If any zone boundaries lie outside of the histogram, that zone's controls will do nothing. In the screenshot above, the zone's range boundaries all lie inside of the visible histogram, so each zone's controls will make a predictable adjustment to the image data in that part of the histogram.



The Zone graph

In the following screenshot, the Zone graph shows that the Specular zone boundary lies outside of the source histogram for the image, so adjustments made using the Specular controls will have no effect. For this reason, it's a good idea to check that the zone boundaries intersect the parts of the image you want to adjust prior to manipulating the controls, but if you notice that a particular zone's controls aren't having as much of an effect on the image as you'd like, you can always move the zone boundary after the fact to expand the range of the image that's affected by that zone.



The Specular zone boundary lies outside of the source histogram, excluding the Specular controls from affecting the image.

Customizing zone boundaries in this way is one of the most powerful aspects of the HDR palette, but don't worry that you'll be constantly dragging zone boundaries back and forth for every shot you grade. Generally speaking, most well-photographed scenes have similar ranges of image tonality in each shot, so zone boundaries that have been customized for one shot from that scene will likely be good for most clips in that scene. That said, adjusting zone boundaries is another powerful way of tailoring a grade specifically for a clip with a unique range of contrast.

Zone Controls

(Min/Max) Range: Defines the level of image tonality at which that zone's adjustment starts. This value is expressed in stops.

Falloff: Adjusts how softly the adjustment in that zone will blend into the image. A value of 0 results in an immediate transition to the adjustment created in the current zone, with the color adjustment being a sharper effect. Higher values will blend the current zone's adjustment into the image more softly, resulting in a more gentle and seamless transition into the effect.

Customizing and Saving Presets

If you've customized the Zone graph in a way that's potentially useful for other grading situations, you can save a preset of the Zone settings for future use. Saving, recalling, and managing these presets is simple.

Methods of managing presets:

- **To save a preset:** Customize the Zone graph to suit your needs, then choose Save As New Preset from the HDR palette Option menu.
- **To load a preset:** Choose Presets > Name of Preset > Load Preset from the HDR palette Option menu. A new preset will be saved.
- **To update a preset:** Load a preset, make any changes you need to from the Zone graph, and then choose Presets > Name of Preset > Update Preset from the HDR palette Option menu.
- **To delete a preset:** Choose Presets > Name of Preset > Delete Preset from the HDR palette Option menu.
- **To select the default preset:** Choose Default Preset > Name of Preset from the HDR palette Option menu. This preset will now be the one loaded initially into the HDR palette on startup.

Resetting Color and Zone Adjustments

In the HDR palette's Option menu, there's a Reset submenu with three commands, Reset All, Reset Color Adjustments, and Reset Zone Definitions, which are self-explanatory. These are important because, during the course of developing a grade, there will be many times when you'll want to reset your color adjustments without changing how you've customized the Zone graph, or vice versa.

Additionally, each of the visible reset buttons in the GUI can be used with keyboard modifiers to specifically target what to reset. The master reset button in the HDR palette's title bar resets the entire palette when clicked (similar to the Reset All command). However, using keyboard modifiers while clicking this button changes what's reset.

- Hold down Command+Shift while clicking the master reset button to reset only the Zone graph's Range and Falloff controls, without resetting your color adjustments.
- Hold down Command+Option while clicking the master reset button to reset only the color adjustments, while leaving the zones alone.

Both of these modifiers also work with the zone-specific reset controls found in the Wheels panel, letting you reset only the Zone controls or only the Color controls of each specific zone.